Communication Systems Lecture-14: Limiters, Mixers and Frequency-Multipliers

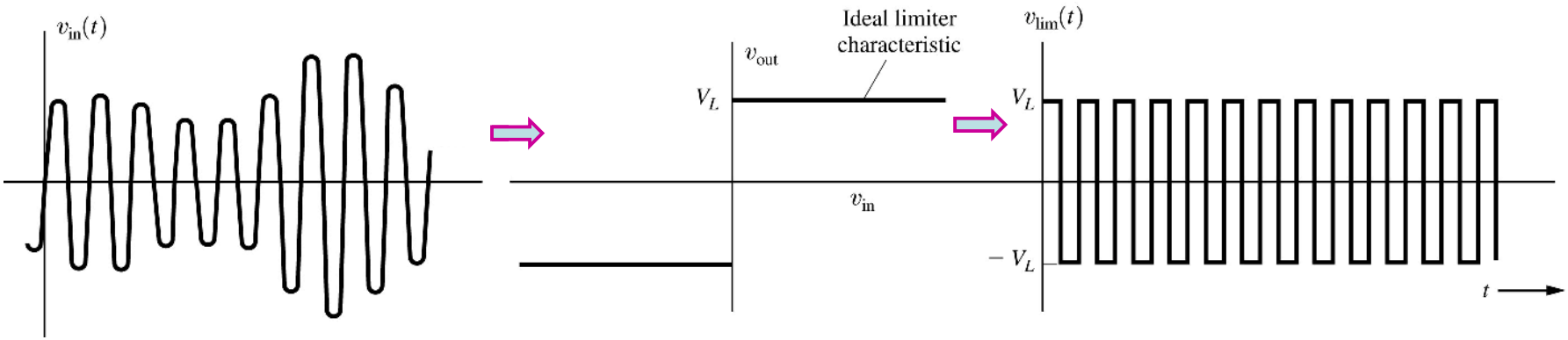
Chadi Abou-Rjeily

Department of Electrical and Computer Engineering
Lebanese American University
chadi.abourjeily@lau.edu.lb

November 1, 2021

Limiters (1)

- An ideal limiter corresponds to a comparator with a zero reference level:
  - When the input voltage takes any positive value, the output of the limiter will be equal to $+V_L$.
  - When the input is negative, the output of the limiter will be equal to $-V_L$.



- The last figure shows how the amplitude variations in the input signal are eliminated in the output signal.

Limiters (2)

- A bandpass limiter corresponds to an ideal limiter followed by a bandpass filter.
- In the case of an ideal bandpass limiter, the output of the filter would be sinusoidal since the harmonics of the square wave would be filtered out.
- In general, any bandpass input can be represented by:

$$v_{in}(t) = R(t)\cos\left[\omega_c t + \theta(t)\right]$$

  - $R(t)$ is the equivalent real envelope.
  - $\theta(t)$ is the equivalent phase function.
- The corresponding output of the ideal bandpass limiter becomes:

$$v_{out}(t) = KV_L\cos\left[\omega_c t + \theta(t)\right]$$

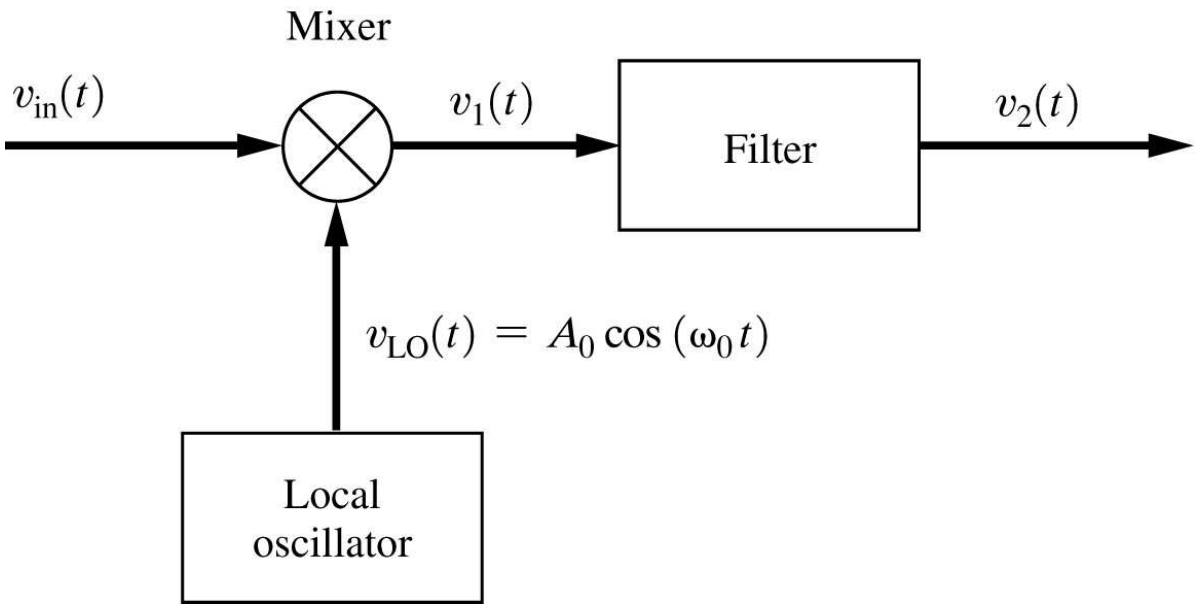
  - K is the level of the fundamental component of the square wave $(4/\pi)$ multiplied by the gain of the bandpass filter.

Limiters (3)

- The last equation shows that any amplitude modulation (AM) that was present on the limiter input does not appear on the limiter output.
- On the other hand, the phase function is preserved. In other words, the zero crossings of the input are preserved on the limiter output.
- Limiters are often used in receiving systems designed for angle-modulated signaling such as PM (phase modulation), FM (frequency modulation), FSK (frequency shift keying) or PSK (phase shift keying). They are used to eliminate any variations in the real envelope of the receiver input signal that are caused by noise.

Mixers (1)

- An ideal mixer corresponds to the multiplication of an input signal with a sinusoidal waveform produced by a local oscillator.
  - The next figure shows a mixer followed by a filter.



- Mixers are often used to obtain a frequency translation of the input signal.

Mixers (2)

- Assume that the input signal is a bandpass signal that has a nonzero spectrum around $f = f_c$. Then this signal can be written as:

$$v_{in}(t) = R(t)\cos\left[\omega_c t + \theta(t)\right]$$

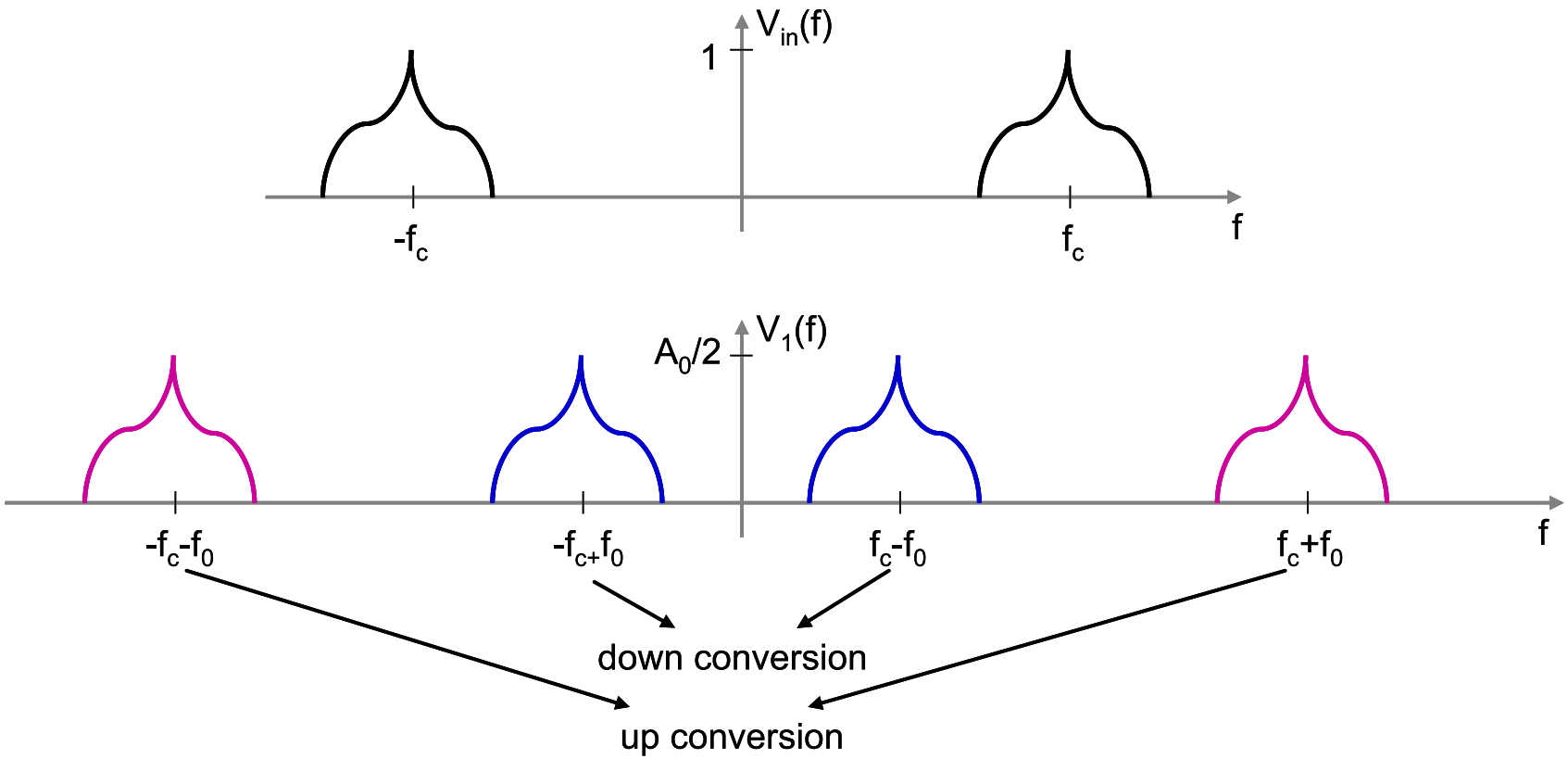
- The signal at the output of the ideal mixer is:

$$\begin{aligned} v_1(t) &= v_{in}(t)\left[A_0\cos(\omega_0 t)\right] \\ &= A_0 R(t)\cos\left[\omega_c t + \theta(t)\right]\cos(\omega_0 t) \\ &= \frac{A_0 R(t)}{2}\cos\left[(\omega_c - \omega_0)t + \theta(t)\right] + \frac{A_0 R(t)}{2}\cos\left[(\omega_c + \omega_0)t + \theta(t)\right] \end{aligned}$$

- The last equation shows that the input signal with a spectrum near $f = f_c$ has been converted to 2 output bandpass signals:
  - An up-conversion band around $f_u = f_c + f_0$.
  - A down-conversion band around $f_d = f_c - f_0$ (we assume that $f_c \geq f_0$).

Mixers (3)

- At the output of the mixer, a bandpass filter is used to select either the up-conversion component or the down-conversion component.
  - Note that when $f_0 = f_c$, a lowpass filter can be used to select the down-converted version of the signal.



Mixers (4)

- $v_1(t)$ can be written as:

$$v_1(t) = \frac{A_0}{2}\mathrm{Re}\left\{g_{in}(t)e^{j(\omega_c+\omega_0)t}\right\} + \frac{A_0}{2}\mathrm{Re}\left\{g_{in}(t)e^{j(\omega_c-\omega_0)t}\right\}$$

  where $g_{in}(t)$ is the complex envelope of $v_{in}(t)$.
- The complex envelope of the signal at the output of the up-converter (located at $f_u = f_c + f_0$) is:

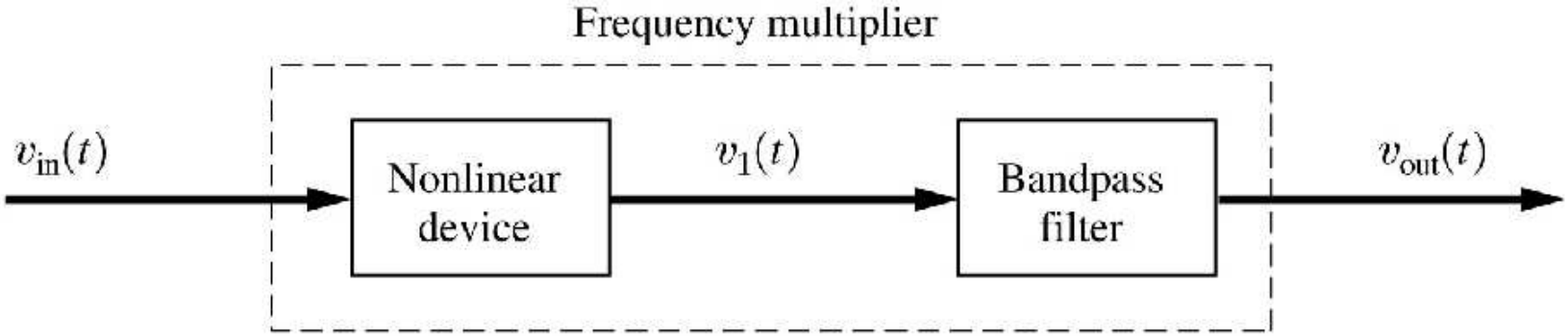
$$g_2(t) = \frac{A_0}{2}g_{in}(t)$$

- In the same way, the complex envelope of the signal at the output of the down-converter (located at $f_d = f_c - f_0$) is:

$$g_2(t) = \frac{A_0}{2}g_{in}(t)$$

- Consequently, up converters and down converters conserve the modulation of their input signals.

Frequency Multipliers (1)



- A Frequency-Multiplier consists of a nonlinear circuit followed by a bandpass filter.
- When the input of a frequency-multiplier is a bandpass signal around $f = f_c$, the output will appear in a frequency band at the n-th harmonic of the input carrier frequency (i.e. around $f = nf_c$).

Frequency Multipliers (2)

- The bandpass input signal can be written as:

$$v_{in}(t) = R(t)\cos\left[\omega_c t + \theta(t)\right]$$

- The output of the nonlinear device is:

$$\begin{aligned} v_1(t) &= K_n v_{in}^n(t) \\ &= K_n R^n(t)\cos^n\left[\omega_c t + \theta(t)\right] \\ &= CR^n(t)\cos\left[n\omega_c t + n\theta(t)\right] + \text{other terms} \end{aligned}$$

  where C is a constant and the other terms of the above equation are separated from the n-th harmonic by at least $2f_c$.

- Using a bandpass filter that passes frequencies in the vicinity of nf_c, the output becomes:

$$v_{out}(t) = CR^n(t)\cos\left[n\omega_c t + n\theta(t)\right]$$

Frequency Multipliers (3)

For example:

- For $n = 2$:

$$\cos^2[\omega_c t + \theta(t)] = \underbrace{\frac{1}{2}}_{\text{DC}} + \frac{1}{2}\underbrace{\cos[2\omega_c t + 2\theta(t)]}_{\text{component at } 2f_c}$$

- For $n = 3$:

$$\begin{aligned}
\cos^3[\omega_c t + \theta(t)] &= \cos^2[\omega_c t + \theta(t)]\cos[\omega_c t + \theta(t)] \\
&= \left[\frac{1}{2} + \frac{1}{2}\cos[2\omega_c t + 2\theta(t)]\right]\cos[\omega_c t + \theta(t)] \\
&= \frac{1}{2}\cos[\omega_c t + \theta(t)] + \frac{1}{4}\cos[3\omega_c t + 3\theta(t)] + \frac{1}{4}\cos[\omega_c t + \theta(t)] \\
&= \frac{3}{4}\underbrace{\cos[\omega_c t + \theta(t)]}_{\text{component at } f_c} + \frac{1}{4}\underbrace{\cos[3\omega_c t + 3\theta(t)]}_{\text{component at } 3f_c}
\end{aligned}$$

- For $n = 4$:

$$\cos^4[\omega_c t + \theta(t)] = \underbrace{\frac{3}{8}}_{\text{DC}} + \frac{1}{2}\underbrace{\cos[2\omega_c t + 2\theta(t)]}_{\text{component at } 2f_c} + \frac{1}{8}\underbrace{\cos[4\omega_c t + 4\theta(t)]}_{\text{component at } 4f_c}$$

Frequency Multipliers (4)

The equation $v_{out}(t) = CR^n(t)\cos[n\omega_c t + n\theta(t)]$ shows that the amplitude variation $R(t)$ is distorted by frequency-multipliers (since the real envelope of the output signal is $R^n(t)$).

- Consequently, frequency-multipliers are not used on signals if the AM is to be preserved.

The waveshape of the angle variation $\theta(t)$ is not distorted by the frequency-multiplier. The magnitude of the angle variation is only multiplied by n.

- Consequently, frequency-multipliers are used with PM and FM since they effectively "amplify" the angle waveform variation $\theta(t)$.

Frequency Multipliers (5)

Frequency multipliers should not be confused with mixers:

- Frequency multipliers are nonlinear devices while mixers perform linear operations.
- The bandwidth of a signal at the output of a frequency multiplier is larger than that of the input signal. On the other hand, the signal at the output of a mixer has the same bandwidth as the input signal.
- For frequency-multipliers, the frequency band of the output is located at the n-th harmonic of the carrier frequency of the input. For mixers, the input spectrum is translated either up or down (depending on the frequency of the local oscillator).